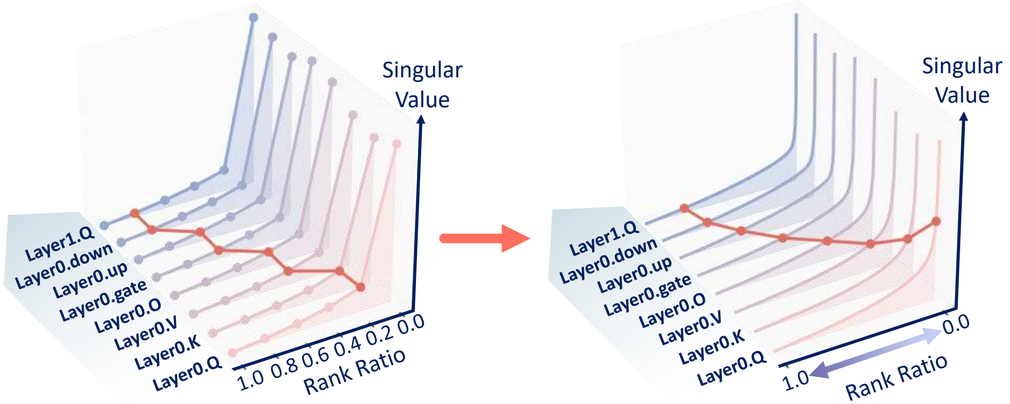
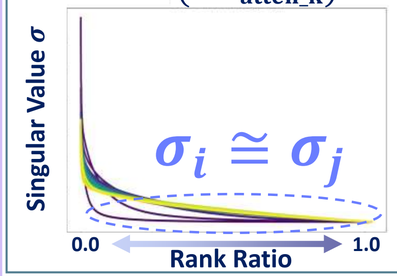
1.Parameter Renormalization: From Discrete to Continuous



2. Robust Backpropagation

Singular Values of Activation

$$\mathrm{SVD}\left(\mathrm{XW}_{\mathrm{atten\_k}}\right)$$



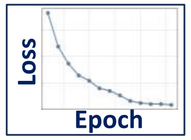
Compute the Gradient of the Loss Function

$$\frac{\partial L}{\partial M} = F\left(\frac{1}{\sigma_i - \sigma_j}, \ldots\right)$$

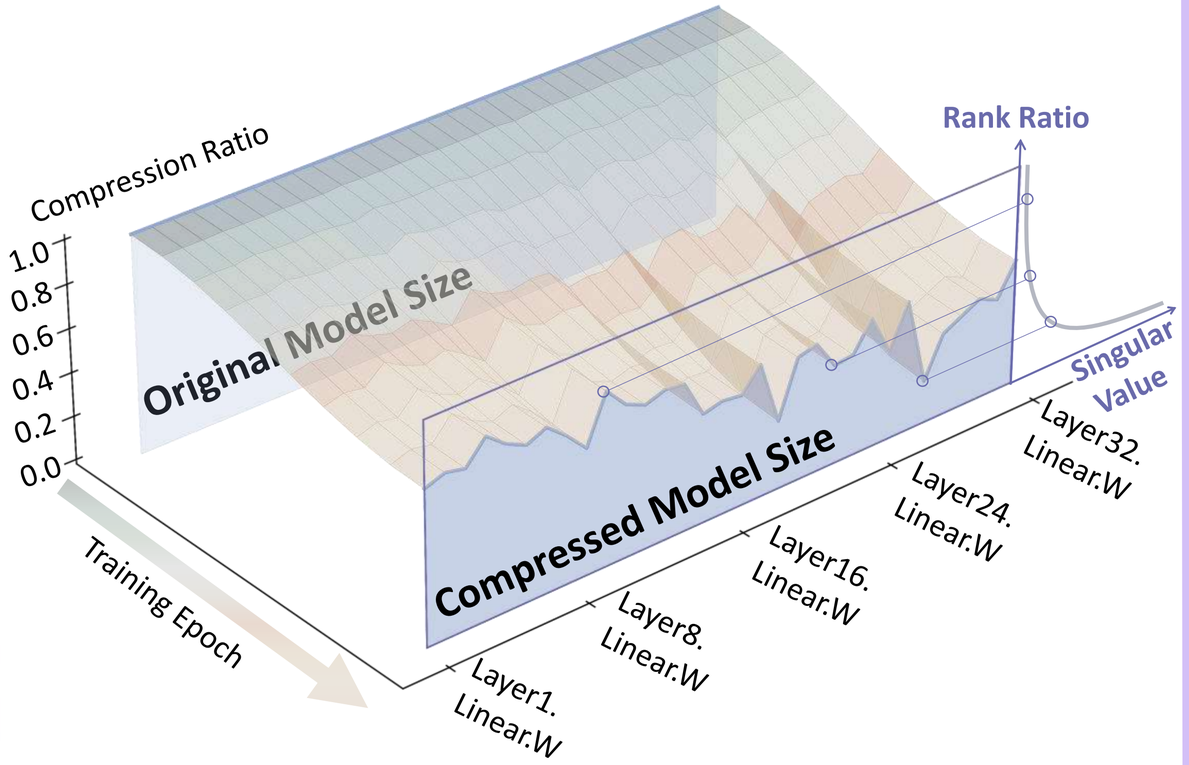
NAN

Taylor Expansion

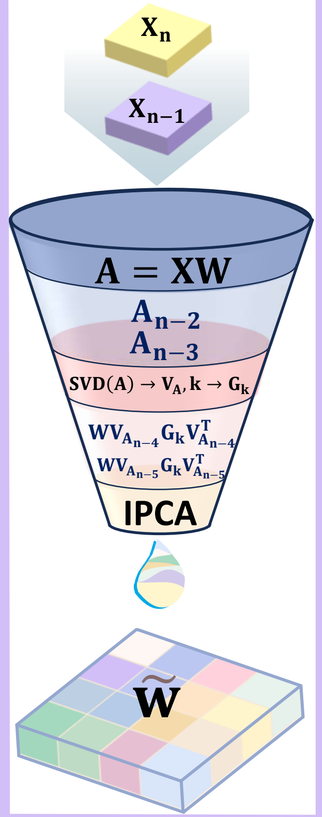
$$\frac{\partial L}{\partial M} = F\left(\frac{1}{\sigma_i}\sum_{n=0}^{K}\left(\frac{\sigma_j}{\sigma_i}\right)^n, \ldots\right)$$



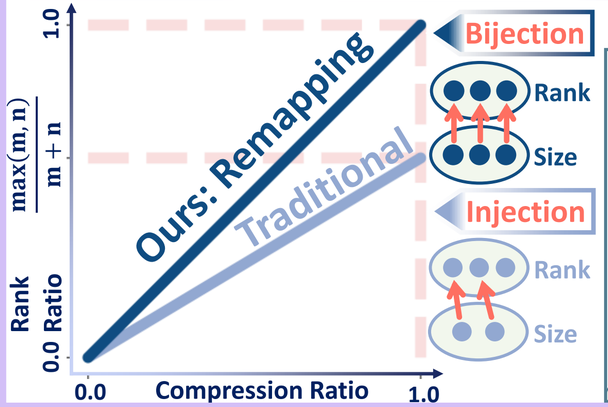
3. Optimization Landscape of Model Compression

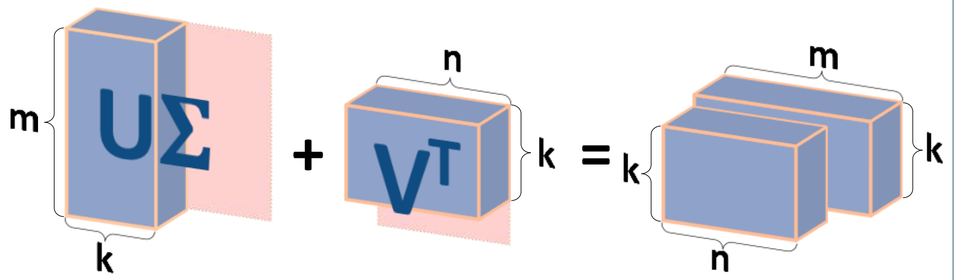


4. Weight Update

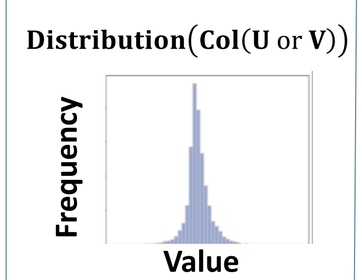


5. Remapping

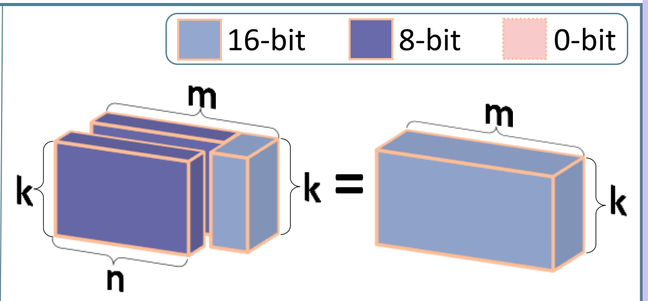




1. Traditional Storage Method: $\mathrm{k} \times (\mathrm{m} + \mathrm{n})$



2.Quantization-friendly



3.Proposed Storage Method:$\mathrm{k} \times \mathrm{m}, \mathrm{m} \geq \mathrm{n}$